

V4.2 Toy Model 3F

Relativistic Phase Constraint and Klein-Gordon / Dirac Bridge

Rev A - Exploratory Mathematical Compatibility Report

Date: 2026.05.25 | Continuation after Toy Model 3E

Status: Toy Model 3F tests whether the phase-geometry track can survive the relativistic energy-momentum constraint. It recovers Klein-Gordon and Dirac structural bridges. It does not derive particle masses, spin values, QFT, or the Standard Model.

1. Purpose

Toy Models 3B to 3E recovered nonrelativistic Schrodinger-type phase dynamics, kinetic structure, scalar potentials, and electromagnetic minimal coupling as hidden phase-geometry structures. Toy Model 3F applies the relativistic stress test.

The nonrelativistic relation $E = p^2 / 2m$ is replaced by the invariant relativistic relation:

$$E^2 = p^2 c^2 + m^2 c^4$$

Term reminders: E is energy, p is momentum, c is the speed of light, and m is rest mass.

2. Klein-Gordon bridge

Use the standard operator substitutions from the phase-wave track:

$$\begin{aligned} E &\rightarrow i \hbar \frac{d}{dt} \\ p &\rightarrow -i \hbar \text{grad} \end{aligned}$$

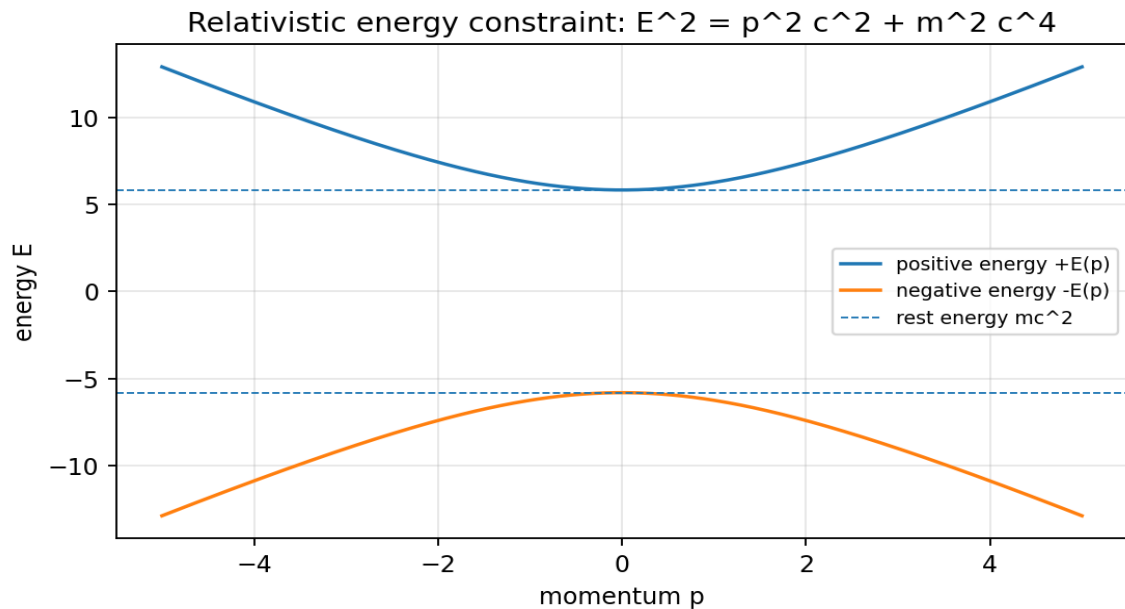
Substituting these into the relativistic invariant gives the Klein-Gordon structural form:

$$\left[\frac{1}{c^2} \frac{d^2}{dt^2} - \text{grad}^2 + (mc/\hbar)^2 \right] \psi = 0$$

For a plane wave $\psi = \exp(i(k \cdot x - \omega t))$, this becomes the dispersion relation:

$$\omega^2 = c^2 k^2 + (mc^2/\hbar)^2$$

Plain meaning: the Klein-Gordon equation is the direct second-order wave equation produced by applying the relativistic energy constraint to the phase-wave substitutions.



3. Why Dirac factorization is needed

The Klein-Gordon equation is second order in time. But Toy Model 3B established first-order time evolution as the clean phase-rotation structure:

$$i \hbar \frac{d\psi}{dt} = H \psi$$

Dirac's move was to factor the relativistic square-root structure into a first-order equation:

$$H = c \alpha \cdot p + \beta m c^2$$

For this to square back to the relativistic invariant, the matrices must satisfy:

$$\alpha_i^2 = I$$

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beta^2 = I
alpha_i alpha_j + alpha_j alpha_i = 0 for i != j
alpha_i beta + beta alpha_i = 0

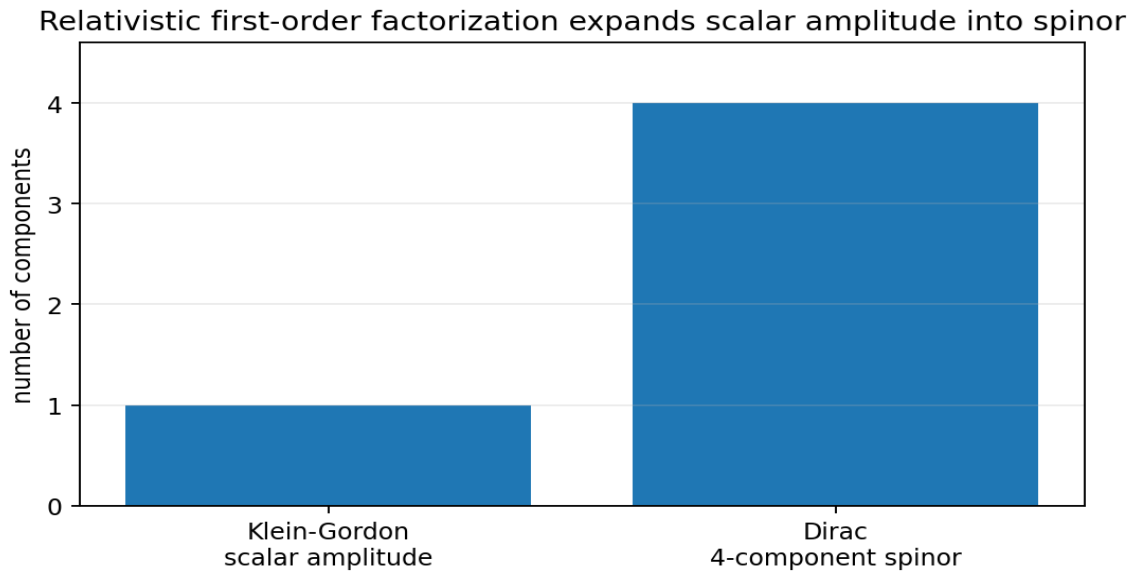
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These are anti-commutation rules. They are the algebraic reason scalar amplitudes are not enough.

4. Emergence of spinor structure

A single scalar number cannot satisfy the needed anti-commuting factorization in three spatial directions plus the mass term. The coefficients must be matrices. Therefore the wavefunction must expand from one complex component into a multi-component object: a spinor.

Term reminder: a spinor is a multi-component quantum object whose components transform under rotations and Lorentz transformations differently from ordinary spatial vectors. In the Dirac equation it carries the structure associated with spin-1/2 and particle/antiparticle branches.



Guardrail: Toy Model 3F does not derive the measured electron spin or particle spectrum. It shows that first-order relativistic factorization forces spinor-like multi-component amplitudes.

5. Dirac equation structural form

With the matrix Hamiltonian, the first-order relativistic equation is:

$$i \hbar \frac{d\psi}{dt} = [c \alpha \cdot (-i \hbar \nabla) + \beta m c^2] \psi$$

Squaring the Hamiltonian gives:

$$H^2 = c^2 p^2 + m^2 c^4$$

This works because the cross terms cancel by anti-commutation. That cancellation is the core algebraic bridge from the Dirac equation back to Klein-Gordon and the relativistic invariant.

6. Gamma matrices and Clifford algebra

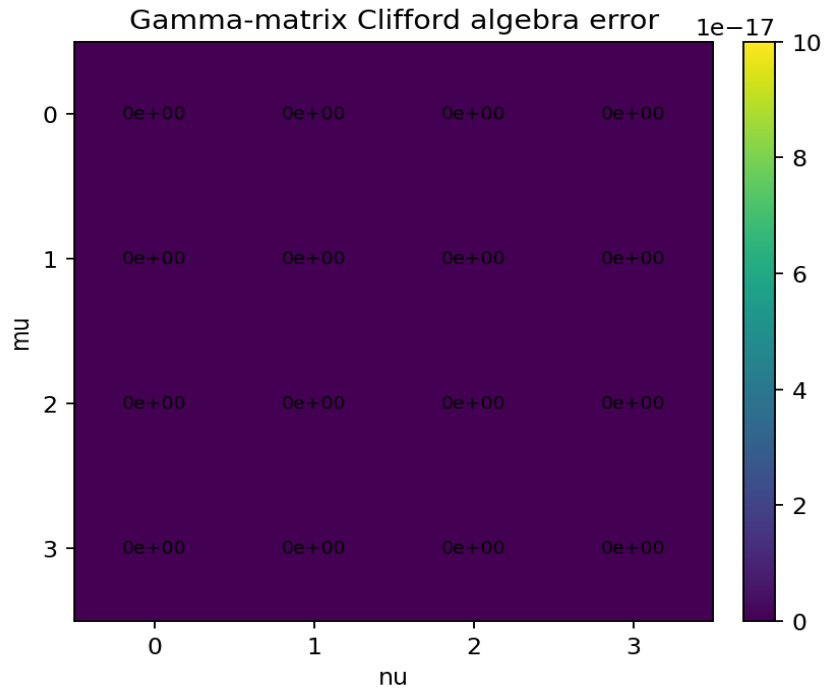
The covariant Dirac form uses gamma matrices satisfying the Clifford algebra:

$$\gamma_\mu \gamma_\nu + \gamma_\nu \gamma_\mu = 2 \eta_{\mu\nu} I$$

Here η is the Minkowski metric. In the sign convention used in this report:

$$\eta = \text{diag}(1, -1, -1, -1)$$

Term reminder: Minkowski metric is the flat spacetime distance rule used in special relativity. Clifford algebra is the anti-commuting matrix algebra that encodes that spacetime signature.

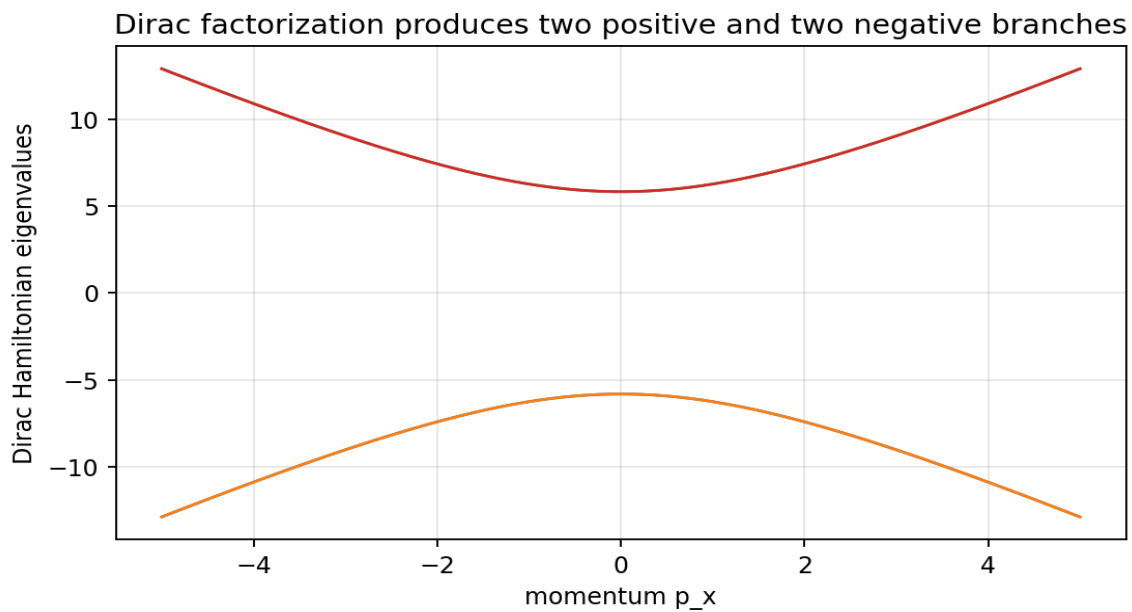


7. Numerical Dirac spectrum check

The Dirac Hamiltonian was tested over random momenta. Its eigenvalues occur as two positive-energy and two negative-energy branches:

$$E = \pm \sqrt{p^2 c^2 + m^2 c^4}$$

The twofold degeneracy is part of the spinor structure. In full physics it is connected with spin degrees of freedom; this toy model only verifies the algebraic structure.



8. Numerical checks

Check	Result
Random samples	10000
Max $\alpha_i^2 - I$ error	0.000e+00
Max α anti-commutator error	0.000e+00
Max α - β anti-commutator error	0.000e+00
Max $\beta^2 - I$ error	0.000e+00
Max gamma Clifford algebra error	0.000e+00
Max Klein-Gordon plane-wave residual	2.220e-14

Check	Result
Max H^2 relativistic invariant error	1.137e-13
Max Dirac eigenvalue error	2.309e-14
Max Dirac plane-wave spinor residual	2.378e-14
Max positive-energy projector error	3.331e-16
Max negative-energy projector error	2.220e-16

9. Interpretation inside the 6D Projection Framework

Toy Model 3F supports the following limited interpretation:

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relativistic phase constraint
-> second-order scalar Klein-Gordon structure
-> first-order time evolution requirement
-> factorization of relativistic invariant
-> anti-commuting matrices required
-> scalar amplitude expands into spinor structure
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In 6D language, the relativistic consistency requirement does not merely adjust phase rotation. It forces the projected amplitude to carry additional internal components. These components are the algebraic shadow of maintaining first-order phase evolution while respecting spacetime invariance.

10. What Toy Model 3F achieves

- It replaces the nonrelativistic dispersion relation with $E^2 = p^2 c^2 + m^2 c^4$.
- It recovers the Klein-Gordon structural equation from the phase-wave operator substitutions.
- It shows why a first-order relativistic equation requires matrix factorization.
- It verifies the Dirac alpha/beta anti-commutation rules.
- It verifies the gamma-matrix Clifford algebra.
- It verifies that the Dirac Hamiltonian squares to the relativistic invariant.
- It shows that spinor-like multi-component amplitudes are structurally required by first-order relativistic factorization.

11. What Toy Model 3F does not achieve

- It does not derive the physical mass of the electron or any particle.
- It does not derive the numerical value of spin or charge.
- It does not derive quantum field theory.
- It does not derive the Standard Model.
- It does not derive interactions beyond structural minimal coupling already tested in 3E.
- It does not prove physical 3T existence.
- It does not generate new experimental predictions.

12. Close-out statement for 3F

Toy Model 3F successfully extends the phase-geometry track into the relativistic domain at the structural level. Applying phase-wave operator substitutions to $E^2 = p^2 c^2 + m^2 c^4$ yields the Klein-Gordon equation. Requiring first-order time evolution forces Dirac-style factorization, which requires anti-commuting matrices and therefore multi-component spinors. The numerical checks confirm the Clifford algebra, the Dirac Hamiltonian square, and the relativistic energy spectrum. This is a structural compatibility result, not a derivation of particle masses, QFT, the Standard Model, or physical 3T existence.

13. Recommended next chapter

The next logical chapter is Toy Model 3G: Quantum Dynamics Close-Out and Guardrail Consolidation.

1. Consolidate Toy Models 3A through 3F into one close-out paper.
2. Separate structural compatibility from derivation.
3. Record the chain from complex amplitudes to Schrodinger structure, potentials, gauge coupling, and relativistic spinor bridge.
4. Identify the remaining hard gaps: Born rule derivation, physical origin of $\hbar$, mass, charge, interaction source equations, QFT, and experimental predictions.

5. Decide whether the next major track should target gravity/dark sector or formal graph-to-metric geometry.

14. Rebind prompt for continuation

We are resuming after Toy Model 3F.

3F tested the relativistic phase constraint and Klein-Gordon / Dirac bridge.

Relativistic invariant:
 $E^2 = p^2 c^2 + m^2 c^4$

Operator substitutions:
 $E \rightarrow i \hbar \frac{d}{dt}$
 $p \rightarrow -i \hbar \text{grad}$

Klein-Gordon structure:
 $[(1/c^2)d^2/dt^2 - \text{grad}^2 + (mc/\hbar)^2] \psi = 0$

First-order Dirac factorization:
 $H = c \alpha \cdot p + \beta m c^2$

Required anti-commutation:
 $\alpha_i^2 = I$
 $\beta^2 = I$
 $\{\alpha_i, \alpha_j\} = 0 \text{ for } i \neq j$
 $\{\alpha_i, \beta\} = 0$

Dirac equation:
 $i \hbar \frac{d\psi}{dt} = [c \alpha \cdot (-i \hbar \text{grad}) + \beta m c^2] \psi$

Result:
First-order relativistic phase evolution forces matrix factorization and multi-component spinor structure.

Guardrail:
3F does not derive particle masses, spin values, charge, QFT, the Standard Model, or physical 3T existence.

Next recommended chapter:
Toy Model 3G - Quantum Dynamics Close-Out and Guardrail Consolidation.